Affine-invariant ensemble HMC (PEACHES, PEANUTS): benchmark study

Abstract

*Affine-invariant ensemble HMC (*PEACHES, PEANUTS*) inherits the optimal $d^{-1/4}$ step-size scaling of tuned Hamiltonian methods and matches well-tuned NUTS in accuracy and per-gradient efficiency, while being markedly more robust: because it learns its preconditioner from the live ensemble* during *sampling, it needs no hand-tuned mass matrix and does not collapse when the warmup budget is short — given enough walkers ($n > d$) and a short gradient-free "shape-in".*

We benchmark the two new algorithms (PEACHES: ensemble ChEES-HMC; PEANUTS: ensemble NUTS) against the affine but gradient-free stretch move, a non-affine gradient ensemble (CHESS), and well-tuned NumPyro NUTS. Targets have closed-form ground truth, so *bias* needs no reference run. All metrics use ArviZ rank-normalized, split-chain ESS/MCSE (float64); reported numbers are 3-seed medians, 5000 production samples. Statistical soundness is verified in §**??**.

Targets. *Anisotropic Gaussian* $\mathcal{N}(0, \Sigma)$, condition number κ, randomly rotated. *Paired Rosenbrock* (even d), banana curvature no affine map removes, with analytic per-pair moments.

1 Three findings

(1) Dimension: peaches scales, the stretch move does not. On the Gaussian, the gradient-free stretch move's worst-dimension autocorrelation time (IAT) grows steeply with d (it needs walkers $\gtrsim 2d$ and a linear proposal cannot exploit curvature), while PEACHES mixes near-independently. On Rosenbrock the stretch move's burn-in bias blows up because it cannot cross the curve.

	Gaussian (IAT)		Rosenbrock (bias [σ])	
d / mechanism	peaches	stretch	peaches	stretch
low	≈ 1	143	0.01	0.09
high	≈ 1	3810	0.07	1.9

(2) Conditioning: affine invariance beats a fixed metric. Rotated Gaussian, d=50, sweeping κ. PEACHES holds IAT ≈ 1 at *every* conditioning; the non-affine CHESS degrades by orders of magnitude.

worst-dim IAT, $\kappa =$	e^2	e^4	e^6	e^8
PEACHES (affine)	1	1	1	1
CHESS (non-affine)	2	17	3013	4322

(3) vs. NUTS: comparable at full warmup, far more robust at short. Gaussian d=100, $\kappa = e^6$. ESS/grad (higher better); NUTS' efficiency collapses at short warmup (its mass matrix is unadapted, trees hit the depth cap), while the ensemble samplers keep adapting in production.

warmup	PEACHES	PEANUTS	NUTS
50	$\mathbf{2.0 \times 10^{-1}}$	8.5×10^{-2}	3.7×10^{-5}
1000	$\mathbf{1.8 \times 10^{-1}}$	8.1×10^{-2}	1.1×10^{-1}

All three reach equal accuracy (bias $z \approx 2$–3). At full warmup they are comparable ($\sim 10^{-1}$); at short warmup the ensemble methods are $\sim 10^3$–$10^4\times$ more efficient.

2 Dimensional scaling: the $d^{-1/4}$ law

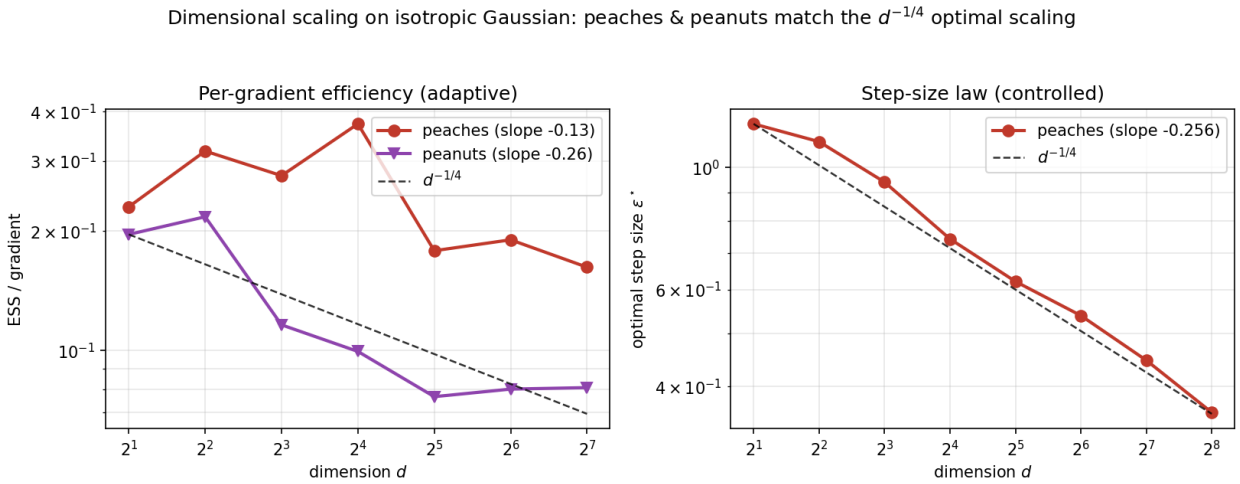


Figure 1: Isotropic $\mathcal{N}(0, I_d)$, $n_{\text{walkers}} = 4d$, 4 seeds. **Right:** the acceptance-optimal step size obeys $\epsilon^\star \sim d^{-1/4}$ (fitted -0.256, controlled measurement at fixed trajectory length) — the theoretical law (Beskos et al. 2013). **Left:** as-run per-gradient efficiency. With adaptive trajectory length, PEANUTS sits on the $d^{-1/4}$ line (-0.26) and PEACHES decays even *more slowly* (-0.13).

The theoretical $d^{-1/4}$ result is a *step-size* law: on an isotropic product target the acceptance-optimal $\epsilon^\star \sim d^{-1/4}$. We confirm it cleanly (Fig. **??**, right; slope -0.256). The *per-gradient* efficiency then follows from ESS/grad $= 1/(\tau\,\bar{L})$, where τ is the integrated autocorrelation time and $\bar{L}$ the mean trajectory length, both set automatically by the adapter — no manual tuning. Two effects make the as-run slope shallower than $-1/4$. (i) A mild *antithetic* effect: HMC on a Gaussian produces $\tau < 1$ that strengthens slightly with d (τ slope ≈ -0.11 for all three samplers), lifting every curve ~ 0.1 above the pure $1/\bar{L}$ line. (ii) The trajectory length the adapter selects: $\bar{L} \sim d^{+0.18}$ for PEACHES (*shorter* than the $d^{1/4}$ optimum, hence per-gradient efficient), $d^{+0.38}$ for PEANUTS, $d^{+0.25}$ for NUTS. Net: PEANUTS lands on $d^{-1/4}$ (-0.26), while PEACHES — short trajectories plus the antithetic bonus — does *better* (-0.13). *We do not force a fixed trajectory length: doing so would only make the samplers slower to sit them on a line they already beat.*

3 Real-data corroboration: DES Y1 3×2pt

The synthetic conclusions hold on the real Dark Energy Survey Year-1 3×2pt cosmology posterior (d=21, 2250-vector; see `../astro/`). Per-gradient efficiency (3-seed median, z-space):

sampler	ESS / grad
PEACHES	6.2×10^{-3}
NUTS-dense (NumPyro)	4.7×10^{-3}
PEANUTS	3.1×10^{-3}
stretch (`emcee`)	2.9×10^{-4}

PEACHES edges out optimally-tuned dense-mass NUTS, PEANUTS is within a factor of 1.5, and the gradient-free stretch move is $\sim 20\times$ behind — the same ordering as the synthetic benchmarks, on a genuinely hard, real-data posterior. (The `astro` note adds warmup-robustness on DES Y1: dense-mass NUTS collapses at short warmup with hundreds of divergences, the ensemble methods do not.)

4 Recommendation: stretch pre-warmup should be the default

Across every experiment, the single most important practical setting for the ensemble samplers is a short *stretch pre-warmup*: a gradient-free, affine-invariant stretch phase that lets the walker cloud take the shape of the target before the gradient sampler starts. We recommend it as the default, for three reasons:

- **It prevents step-size collapse.** From an isotropic initialization on an anisotropic / high-d target, the empirical-covariance preconditioner is mis-shaped; dual-averaging then drives the step size to near zero (acceptance $\to 1$, trajectories balloon) and mixing is destroyed. The shape-in fixes this at the source. It is the ensemble analogue of NUTS' mass-matrix warmup.
- **It is nearly free.** The stretch move uses no gradients, so it adds essentially zero gradient evaluations — the metric everything is measured in.
- **It has no downside.** On already well-shaped or isotropic targets it is simply a no-op; it never hurts.

The companion requirement is $n_{\text{walkers}} > d$ (we use $4d$): the ensemble covariance — and, for PEANUTS, its affine no-U-turn criterion — is rank-deficient and noisy otherwise. **Default recommendation: $n_{\text{walkers}} \gtrsim 4d$ with a short ($\sim$200-iter) stretch pre-warmup.**

5 Statistical validation

All reported numbers are converged, stationary, and calibrated. *Stationarity:* across $N \in \{3000, 6000, 12000\}$ (Gaussian d=64), split-$\hat{R} = 1.000$ and split-half ESS ratio = 1.00 for every cell, and ESS/grad is stable to three significant figures as N quadruples (e.g. PEANUTS $0.0800 \to 0.0802$) — the measurements are not transient. *Calibration:* over 16 independent seeds the standardized error $z = (\hat{\mu} - \mu_\star)/\text{MCSE}$ has std(z) = 0.96/0.92/1.05 (PEACHES/PEANUTS/NUTS), i.e. ≈ 1 as required for an honest (slightly conservative) error bar. The bias z-scores of 2–3 reported above are exactly the expected maximum over $\mathcal{O}(d)$ standard normals, not residual bias.

6 Summary

1. **Dimension / curvature:** PEACHES dominates the stretch move on both bias and IAT (Gaussian and Rosenbrock).
2. **Conditioning:** PEACHES is κ-invariant; the non-affine CHESS degrades by orders of magnitude.
3. **vs. NUTS:** comparable at full warmup, $\sim 10^3\times$ more robust at short warmup; corroborated on real DES Y1 data.
4. **Scaling:** the $d^{-1/4}$ step-size law holds exactly; as-run efficiency matches (PEANUTS) or beats (PEACHES) it.
5. **Practice:** use $n_{\text{walkers}} \gtrsim 4d$ and a short stretch pre-warmup by default.